Introduction to Data Science Assignment 1

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Content

Part 1 - Data Exploration 2
- a) 2
- b) 2
- c) 2
- d) 2
- e) 3
- f) 3
- g) 4
- h) 4
- i) 5
- j) 5

Part 2 - Decision Trees 5
- a) 5
- b) 5
- c) 6
- d) 6

Part 3 – Clustering 7
- a) 7

Part 4 – Regression 8
- a) 8
- b) 9

Part 5 – SVM 9
- a) 9
- b) 9
- c) 10
- d) 10

Part 6 – Neural Networks & Naïve Bayes 12
- a) 12
- b) 13
- c) 13

Part 1 - Data Exploration

a)

The dataset has 7 columns and 1349 rows.

b)

	age	bmi	children	charges
count	1348.000000	1347.000000	1348.000000	1347.000000
mean	39.228487	30.655499	1.103858	13254.716622
std	14.063585	6.085427	1.217132	12096.109347
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.315000	0.000000	4742.306100
50%	39.000000	30.360000	1.000000	9377.904700
75%	51.000000	34.637500	2.000000	16582.138605
max	64.000000	53.130000	7.000000	63770.428010

1. Overview of the basic statistics of insurance_orig

Other features are categorical attributes. The method describe() only includes all numeric columns by default.

c)

The dataframe, which contains only rows without any NaN values, has 1338 rows.

d)

Number of underweight: 20.

Number of normal weight: 225.

Number of overweights: 386.

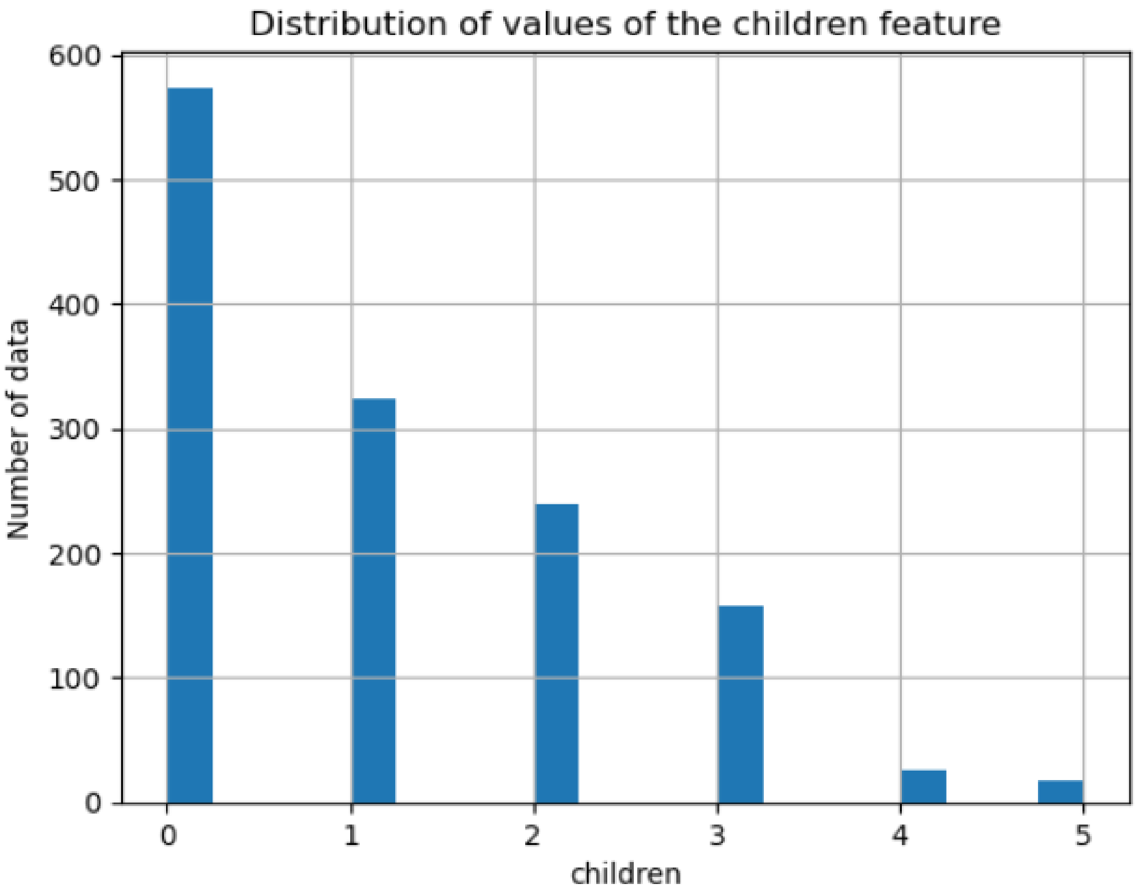
Number of obese: 707.

The dataset is not well balanced. For example, the difference between the number of underweight and obese samples is very large. A model trained on such an imbalanced dataset will likely achieve high accuracy when predicting the obese class but poor accuracy when predicting the underweight class.

Under-sampling and Over-sampling could be used to balance out their representation. With under-sampling we leave out instances of the over-represented group and create balance between groups.

For over-sampling we duplicate instances of the under-represented group to create balance between groups.

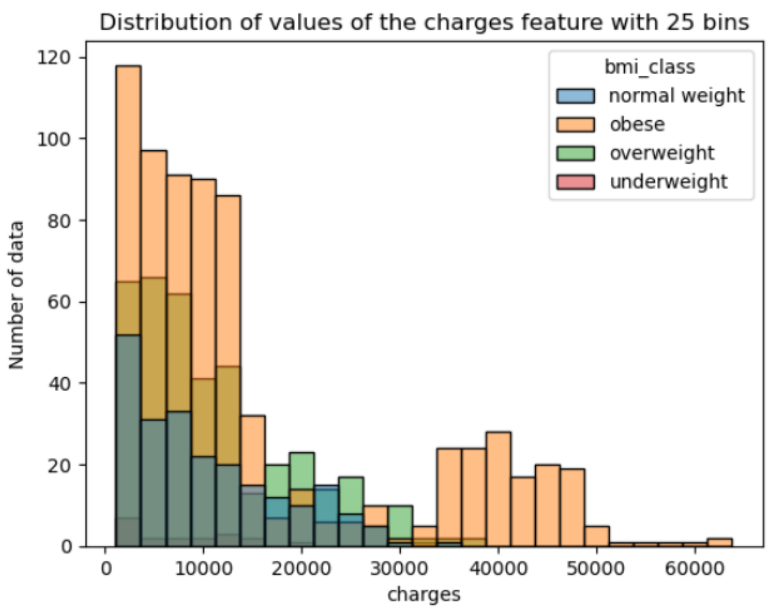
e)



2. Distribution of values of the children feature

The most common value of the children feature is 0, so the mode of the children feature is 0.

f)



3. Distribution of values of the charges feature with 25 bins

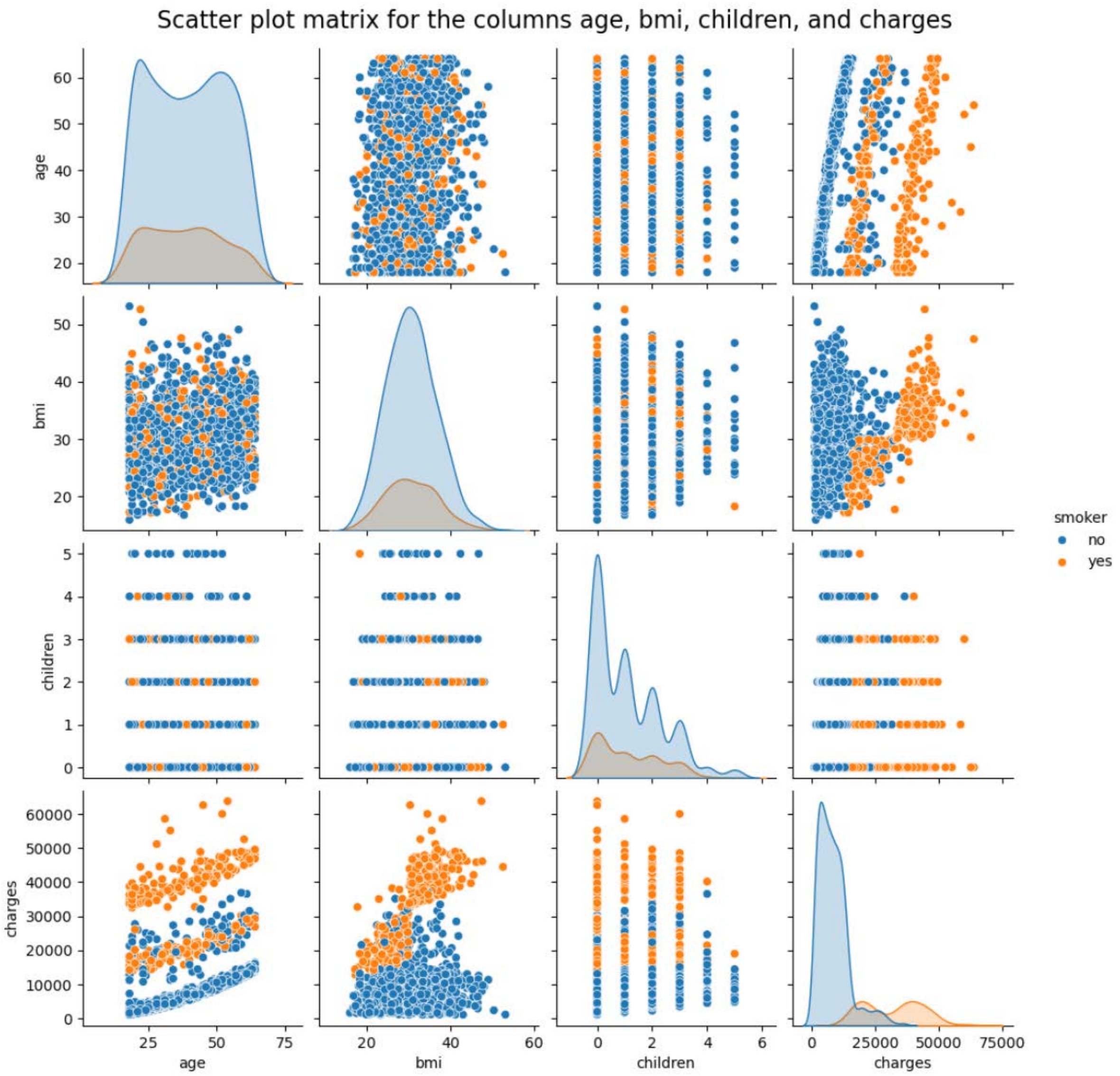
The number of bins is 25. The course of the distribution can be observed well while the effects of minor observation errors are reduced.

The histogram follows an exponential-like distribution, with its peak occurring between 0 and 10,000.

g)

At lower charges, between 0 and 30,000, the BMI classes are well distributed. However, above 30,000, the obese BMI class completely dominates the distribution, and the other classes are barely visible. This indicates a trend: the higher the charges, the more likely it is that the patient has a high BMI.

h)



4. Scatter plot matrix for the columns age, bmi, children, and charges

At lower charges, all the patients are non-smokers, which appears to be independent of other features (age, BMI, and number of children).

i)

	age	bmi	children	charges
age	1.000000	0.109272	0.042469	0.299008
bmi	0.109272	1.000000	0.012759	0.198341
children	0.042469	0.012759	1.000000	0.067998
charges	0.299008	0.198341	0.067998	1.000000

5. Correlations for all the feature pairs from age, bmi, children, and charges

The strongest absolute correlation is 0.30 and it is the correlation between charges and age. This indicates that the older the person is, the higher the medical insurance cost of the individual.

j)

1. There are more smokers with charges above 20,000.

2. No, there is no value below the lower fence value.

3. The question cannot be answered. The boxplot of the charges grouped by smokers and non-smokers among the individuals does not provide the mean.

4. Yes, the lower 1st quartile fence of smokers is lower than the upper whisker of the non-smokers

Part 2 - Decision Trees

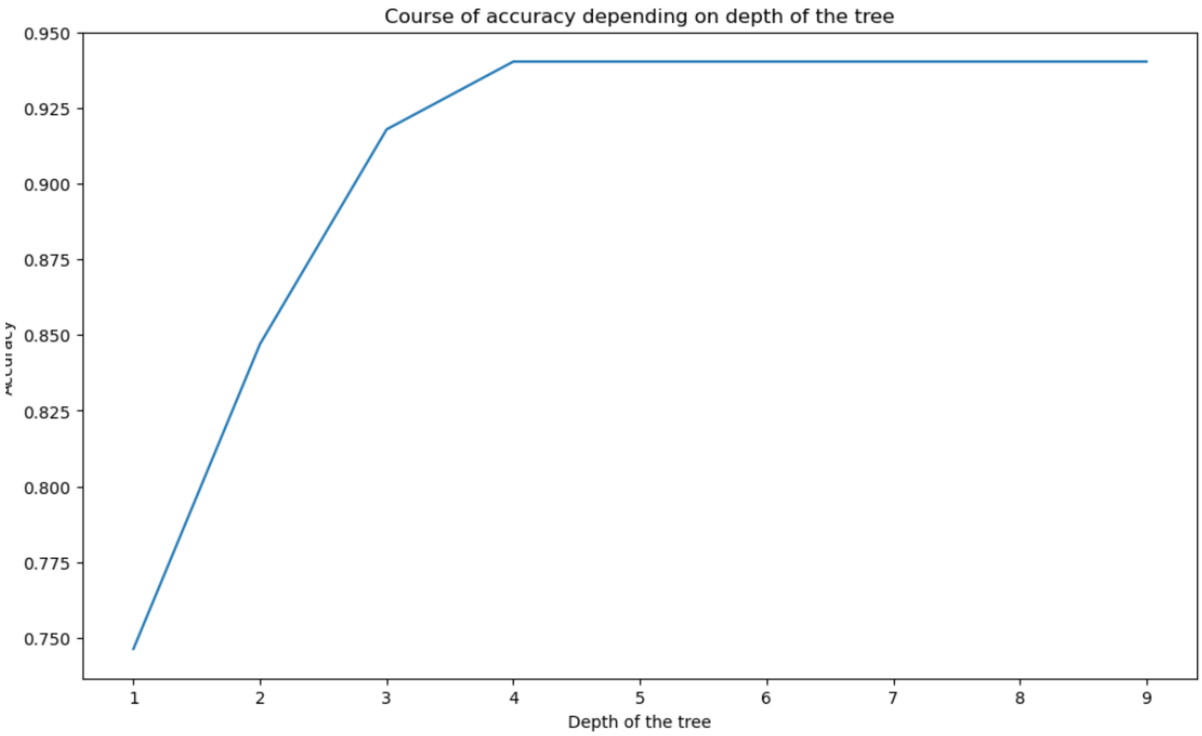
a)

The accuracy of prediction is around 0.49.

b)

The most common value of the chargeGroup feature is medium, so the mode of the chargeGroup feature is medium. The prediction accuracy is around 0.39, which is lower than the accuracy of the wishful-thinking model. This shows that the distribution of one dataset does not necessarily match the distribution of another, and that the mode of the training data is static. A different dataset may have a different dominant class. For example, in the test dataset, the most common value of the chargeGroup feature is *low*, which results in a lower accuracy.

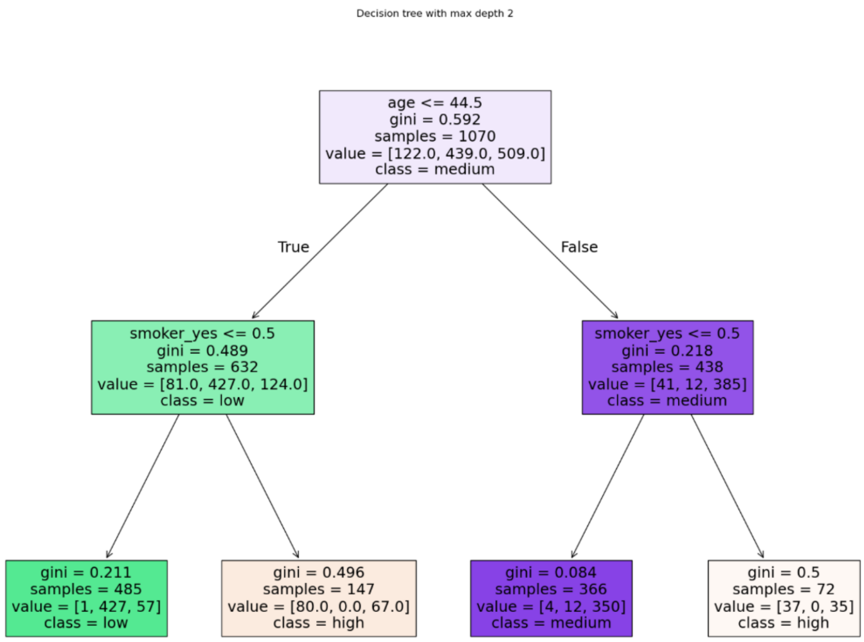
c)



6. Course of accuracy depending on the depth of the tree

We would choose a maximum depth level of 4 because the accuracy is highest at this level. Although the accuracy remains the same for depths above 4, the model will be more complex and performance will decrease. The highest accuracy is around 0.94.

d)



7. Decision tree with max depth 2

A person younger than 45 and non-smoker would lead to a classification of the chargeGroup as medium in the leaf nodes of the decision tree.

The model predicts that the person belongs to the low charge group.

Part 3 – Clustering

a)

The centroids of clusters are

['-0.84', '-0.30', '-0.53', '-0.58']

['0.83', '0.34', '-0.46', '0.64']

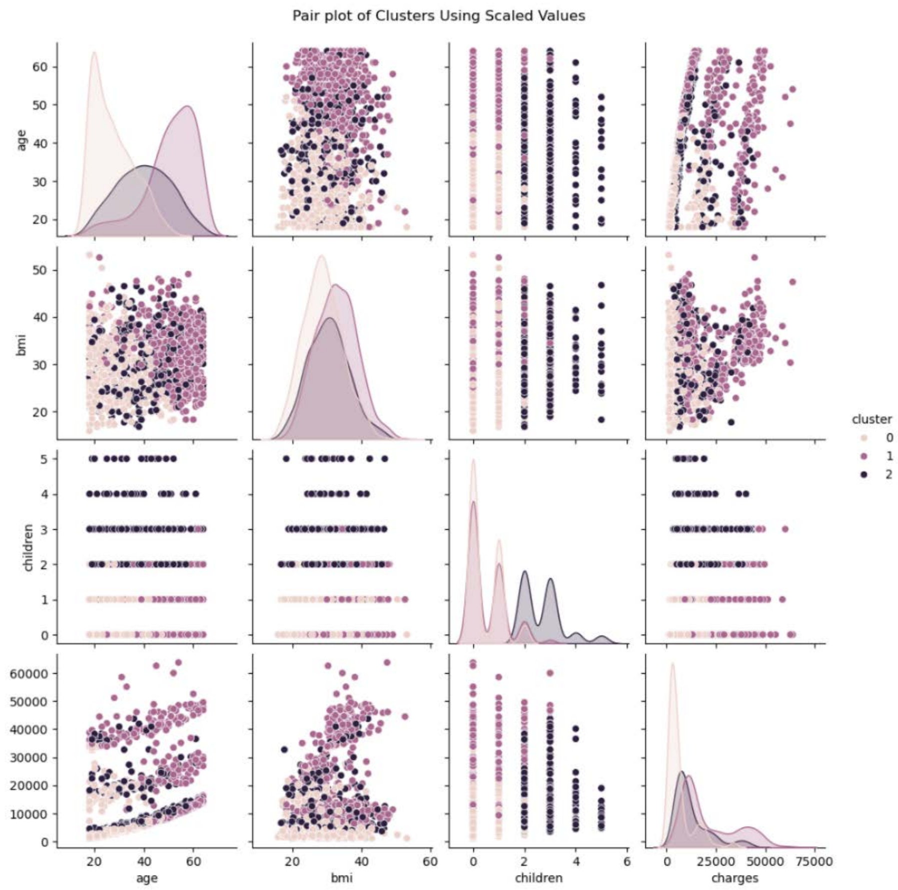
['0.06', '-0.03', '1.34', '-0.05']

The sizes of the clusters are: 498, 480, 360.

The first cluster has 443 non-smokers and 55 smokers.

The second cluster has 323 non-smokers and 157 smokers.

The third cluster has 298 non-smokers and 62 smokers.

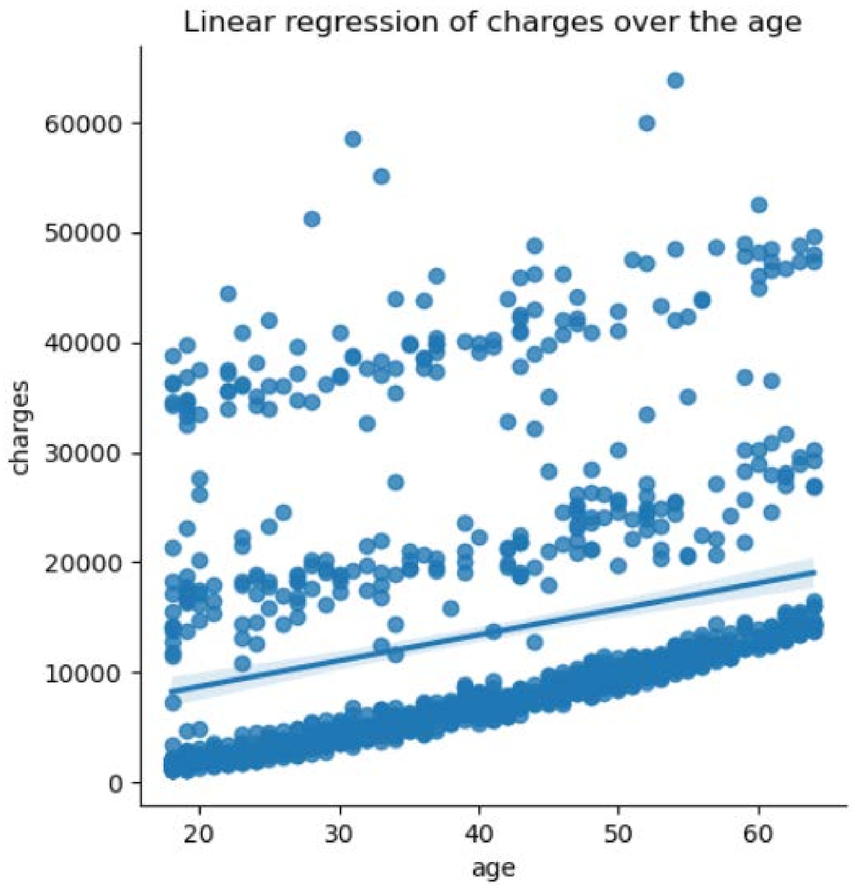


8. Pair plot of clusters using scaled values

The pairplot indicates some separation between clusters in plots involving age. Person with low age and low charges tend to form a distinct cluster, while older people with high charges often belong to another cluster. Young people with 0–1 children form one cluster, and older person with 0–1 children form another. A third cluster consists of people with more than two children, regardless of age. BMI appears to have very little impact on the formation of the clusters.

Part 4 – Regression

a)



9. Linear regression of charges over the age

The plot shows that the linear regression line is nearly horizontal. On one side, the points are densely packed along the x-axis, while on the other side, they are widely scattered. This indicates that a simple linear regression using age alone is a poor fit and that the relationship between age and charges is weak.

b)

The coefficients of the model are:

['257.53', '354.41', '376.80', '38.44', '-38.44', '-11793.39', '11793.39', '622.32', '175.45', '-536.06', '-261.71']

The Mean Absolute Error (MAE) of the training set is 4186.98.

The Mean Absolute Error (MAE) of the test set is 4306.85.

Part 5 – SVM

a)

Confusion Matrix of the first SVM:

[[220   3]

 [9 36]]

Confusion Matrix of the second SVM:

[[221   2]

 [9 36]]

b)

Accuracy of the first SVM:0.96

Accuracy of the second SVM:0.96

Precision of the first SVM:0.92

Precision of the second SVM:0.95

Both the accuracy and precision of the SVM using the three features are very high. This indicates that it correctly predicts and classifies most of the samples, and the model is well suited to the task. The second SVM has the same accuracy as the first SVM but higher precision, which matches expectations. Since the expensive feature depends on charges, the medical insurance cost of an individual, it is natural that the cost has a strong relationship with age, body mass index, and smoking status, and a weaker relationship with the person's gender and region. This explains the accuracy observed. Using more features in second SVM leads to higher precision, because it allows the SVM to better distinguish between classes, particularly the positive class.

c)

Elapsed time of first svm with scaled data: 0.0232 seconds

Elapsed time of second svm with scaled data: 0.0394 seconds

Elapsed time of first svm without scaling data: 0.3539 seconds

Elapsed time of second svm without scaling data: 0.4600 seconds

Confusion Matrix of the first SVM scaled:

[[220   3]

 [9 36]]

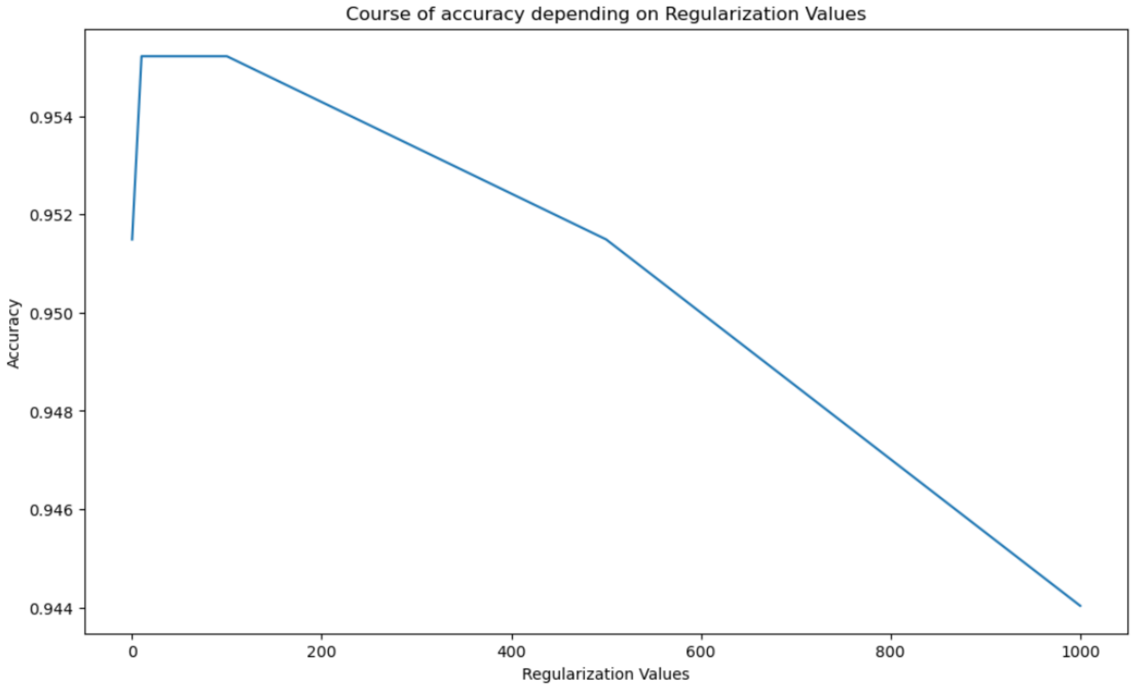
Confusion Matrix of the second SVM scaled:

[[220   3]

 [9 36]]

No improvement is observed after scaling. The second SVM shows lower precision than before. Overall, both SVMs perform equally well. With scaled data, both SVMs require less time to run the pipeline.

d)



10. Course of accuracy depending on regularization values

Among the tested regularization values (1000, 500, 100, 50, 10, 0.1), the highest accuracy of approximately 0.96 is obtained when the regularization parameter is set to 100, 50, or 10. With decreasing regularization values, the accuracy initially increases, but when the value becomes very small, for example C = 0.1, the accuracy drops again. A high regularization value forces the model to fit the training data very strictly, increasing the risk of overfitting because the model tries to classify every data point correctly. This often leads to lower test accuracy. As the regularization value decreases, the model becomes less complex and allows more classification errors, which can improve generalization. However, if the value becomes too small, such as C = 0.1, the model tolerates too many errors and underfits, causing accuracy to decrease again.

e)

Normalization makes features comparable by putting them on a similar scale. Without it, large-scale features, for example charges ranging up to 50,000 can dominate the model and affect accuracy. In the SVM above, the features already had similar ranges or were binary, therefore, normalization had no effect on accuracy.

Part 6 – Neural Networks & Naïve Bayes

a)

Confusion Matrix of MLP:

[[31   0   2]

 [0 131   0]

 [7 12 85]]

Accuracy of MLP: 0.92

Number of instances with probability above 90%: 195

Number of total instances: 268

Class_variance of MLP: ['0.08', '0.20', '0.15']

Probability estimates for the first five instances of the test set:

['0.00', '0.93', '0.07']

['0.00', '0.93', '0.07']

['0.00', '0.92', '0.08']

['0.00', '0.91', '0.09']

['0.02', '0.00', '0.98']

The confusion matrix shows that the second class was predicted perfectly, with 131 true positives. Other two classes have some combined misclassifications, but considering the total number of instances, the model still performs very good. This indicates that the charge group can be reliably predicted with these features. Most instances (195 out of 268) have predicted probabilities above 90% for a certain class, and the variances are very low, further emphasizing that classes are well separated.

b)

Confusion Matrix of MLP:

[[32   0   1]

 [0 131   0]

 [4 12 88]]

Accuracy of MLP: 0.94

One approach to improve the model's performance is feature scaling. For training data set, we normalized the features using sklearn's StandardScaler and retrained the model on the scaled data. We also performed some hyperparameter tuning. Since the dataset contains only 268 entries, a deep network is unnecessary and can lead to overfitting, so we reduced hidden layer size to 2×5. Using this combined approach, the model's accuracy improved, reaching approximately 94%.

c)

Confusion Matrix of GaussianNB without scaling:

[[33   0   0]

 [0 131   0]

 [21 82   1]]

Accuracy of GaussianNB without scaling: 0.62

Confusion Matrix of GaussianNB with scaling:

[[33   0   0]

 [0 131   0]

 [21 82   1]]

Accuracy of GaussianNB with scaling: 0.62